Memory

FMC Bus length matching within
±150 mil (3.8 mm) across the entire bus

STM32H7S3I8T6

Interface_Headers

SDRAM_A0
SDRAM_A1
SDRAM_A2
SDRAM_A3
SDRAM_A4
SDRAM_A5
SDRAM_A6
SDRAM_A7
SDRAM_A8
SDRAM_A9
SDRAM_A10
SDRAM_A11
SDRAM_A12
FMC_A0
FMC_A1
FMC_A2
FMC_A3
FMC_A4
FMC_A5
FMC_A6
FMC_A7
FMC_A8
FMC_A9
FMC_A10
FMC_A11
FMC_A12
Length Match
FMC_A[0..14]

SDRAM_DQ0
SDRAM_DQ1
SDRAM_DQ2
SDRAM_DQ3
SDRAM_DQ4
SDRAM_DQ5
SDRAM_DQ6
SDRAM_DQ7
NAND_D0
NAND_D1
NAND_D2
NAND_D3
NAND_D4
NAND_D5
NAND_D6
NAND_D7
FMC_D0
FMC_D1
FMC_D2
FMC_D3
FMC_D4
FMC_D5
FMC_D6
FMC_D7
FMC_D8
FMC_D9
FMC_D10
FMC_D11
FMC_D12
FMC_D13
FMC_D14
FMC_D15
Length Match
FMC_D[0..15]

SDRAM_BA0
SDRAM_BA1
SDRAM_CLK
SDRAM_CKE
SDRAM_CS
SDRAM_RAS
SDRAM_CAS
SDRAM_WE
SDRAM_DQMH
FMC_BA0
FMC_BA1
PD9_FMC_SDCLK
PE15_FMC_SDCKE0
PE14_FMC_SDNE0
PE12_FMC_SDNRAS
PE13_FMC_SDNCAS
PE11_FMC_SDNWE
PD8_FMC_NBL0

NAND_WP#
NAND_CLE
NAND_ALE
NAND_WE#
NAND_R/B#
NAND_CE#
NAND_RE#
PC13
FMC_CLE
FMC_ALE
FMC_NWE
PD13_FMC_INT
FMC_NCE
FMC_NOE

PD6_FMC_NE3

PM11
PF9_TIM14_CH1
PD15
PO4
PP4
PO1
PP1
PO0
PB11
PM14
PM13
PF4
PC0
PM12
PP3
PP2
PP0
PO5
PF10
PM3
PD2
PM8

File: Memory.kicad_sch

LCD_Interface

FMC_D0
FMC_D1
FMC_D2
FMC_D3
FMC_D4
FMC_D5
FMC_D6
FMC_D7
FMC_D8
FMC_D9
FMC_D10
FMC_D11
FMC_D12
FMC_D13
FMC_D14
FMC_D15
FMC_A9
FMC_NWE
FMC_NOE
FMC_NE3
BTB_LCD_40
BTB_LCD_37
BTB_LCD_43
LED_Driver_CTRL
LCD_TE
LCD_RESET

PA6_TIM13_CH1
PB0_GFXTIM_TE

File: LCD.kicad_sch

PB7_PSSI_RDY
PF8_PSSI_PDCK
PD4_PSSI_DE
PC6_PSSI_D0
PC7_PSSI_D1
PC8_PSSI_D2
PC9_PSSI_D3
PD10_PSSI_D4
PD3_PSSI_D5
PD11_PSSI_D6
PD14_PSSI_D7

PB5_SPI3_MOSI
PC11_SPI3_MISO
PC10_SPI3_SCK
PD12

PB15_SPI2_MOSI
PB14_SPI2_MISO
PB10_SPI2_SCK
PB12_SPI2_NSS

I2S RX
PA5_I2S1_CK
PB4_I2S1_SDI
PA4_I2S1_WS

I2S TX
PA3_I2S6_MCKD
PC12_I2S6_CK
PA15_I2S6_WS
PA7_I2S6_SDO

PB6_LPUART1_TX
PF7_UART7_TX
PB13_UART5_TX
PB9_I2C1_SDA
PB8_I2C1_SCL

PF11_ADC1_INP2
PP7
PO2
PP6
PP5
BOOT0
PM9
PM2

USB_OTG_HS_DP
USB_OTG_HS_DM

PM0_UCPD1_CC1
PM1_UCPD1_CC2

PA14_DEBUG_JTCK-SWCLK
PA13_DEBUG_JTMS-SWDIO
PB3_DEBUG_JTDO-SWO
NRST

File: STM32H7S3I8T6.kicad_sch

BTB43
BTB31
BTB53
BTB11
BTB13
BTB14
BTB16
BTB7
BTB55
BTB8
BTB10

BTB46
BTB37
BTB40

BTB5
BTB4
BTB17
BTB1

BTB22
BTB47
BTB20

BTB19
BTB38
BTB41
BTB30

BTB44
BTB32
BTB2
BTB34
BTB35

BTB23
BTB25
BTB26
BTB28
BTB29
BTB49
BTB50
BTB52

SI5351_I2C_SCL
SI5351_I2C_SDA
SI5351_CLK0

USB_DI_D2+
USB_DI_D2-

RST_BTN

SWCLK
SWDIO
SWO
NRST

BTB3
BTB6
BTB9
BTB12
BTB15
BTB18
BTB21
BTB24
BTB27
BTB33
BTB36
BTB39
BTB42
BTB45
BTB48
BTB51
BTB54
BTB57
BTB56

BTB59
BTB60
BTB58
SMPS_OUT2_3V3+

File: Interface_Headers.kicad_sch

Power_Management

RST_LOCAL

File: Power_Management.kicad_sch

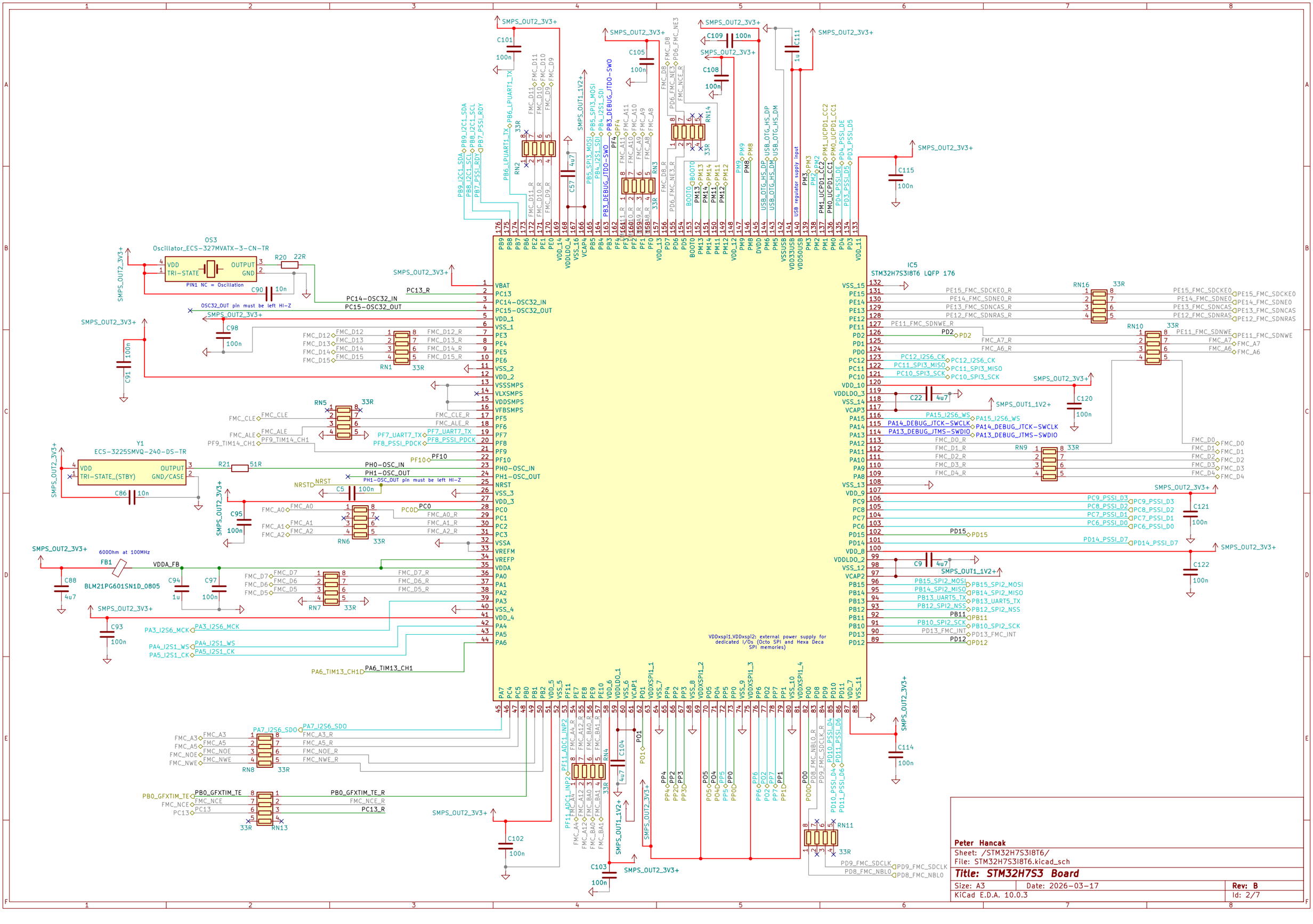
Notes

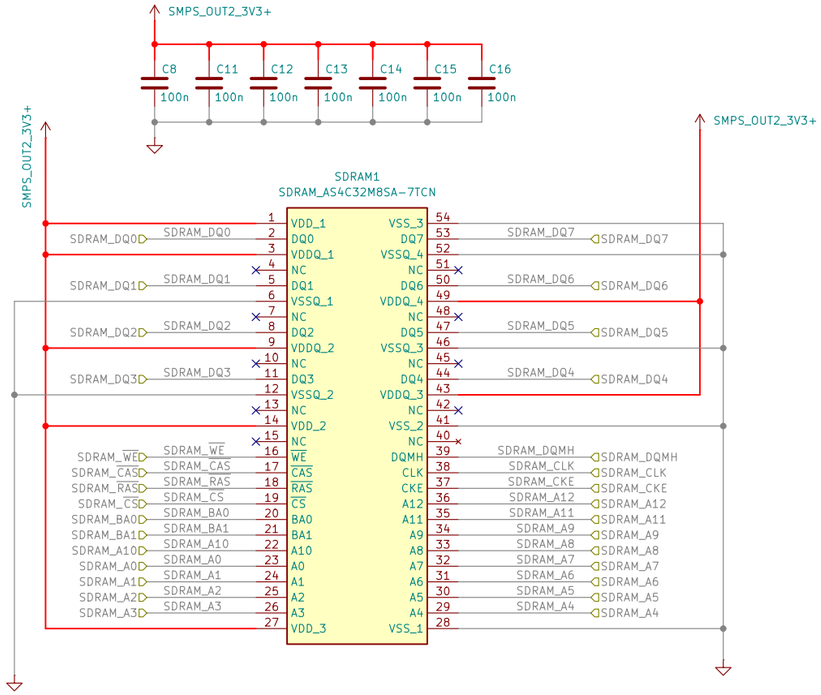
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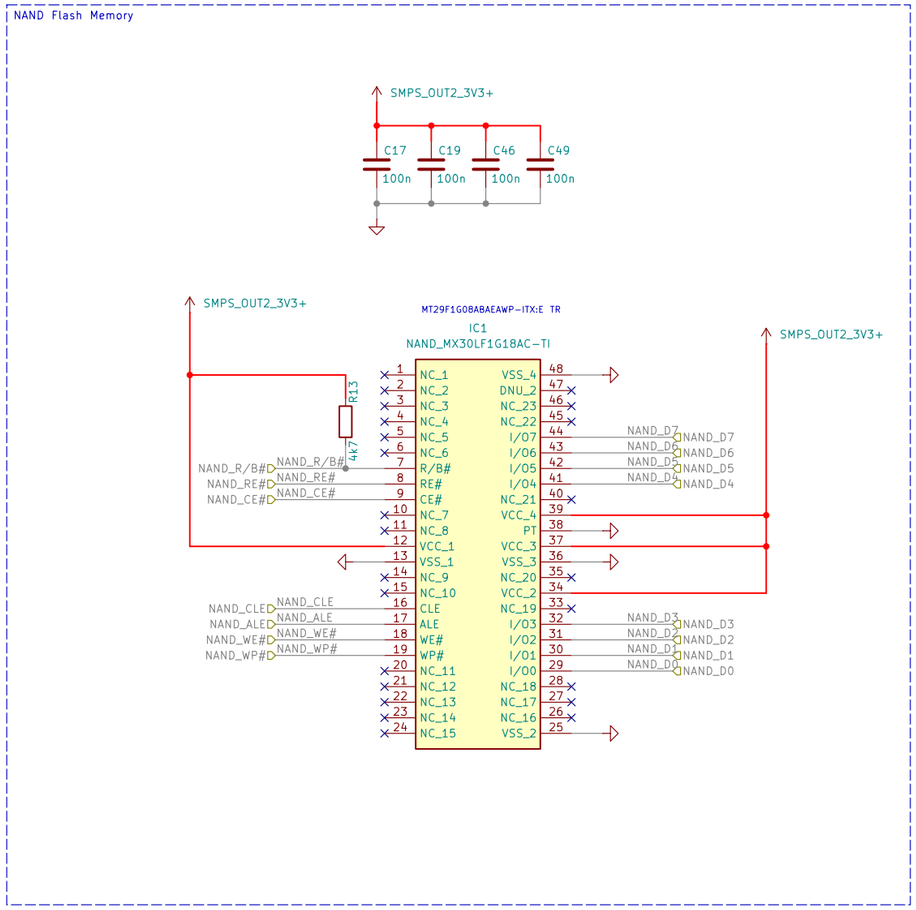
Peter Hancak
Sheet: /
File: STM32H7_Board.kicad_sch

Title: STM32H7S3 Board

Size: A3	Date: 2026-03-17	Rev: B
KiCad E.D.A. 10.0.3		Id: 1/7


IC5
STM32H7S3I8T6 LQFP 176
OS3
Oscillator_ECS-327MVATX-3-CN-TR
PIN1 NC = Oscillation
OSC32_OUT pin must be left HI-Z
Y1
ECS-3225SMVQ-240-DS-TR
PH1-OSC_OUT pin must be left HI-Z
600Ohm at 100MHz
BLM21PG601SN1D_0805
USB regulator supply input
VDDxspi1,VDDxspi2: external power supply for dedicated I/Os (Octo SPI and Hexa Deca SPI memories)
SMPS_OUT2_3V3+
SMPS_OUT1_1V2+
Peter Hancak
Sheet: /STM32H7S3I8T6/
File: STM32H7S3I8T6.kicad_sch
Title: STM32H7S3 Board
Size: A3
Date: 2026-03-17
KiCad E.D.A. 10.0.3
Rev: B
Id: 2/7





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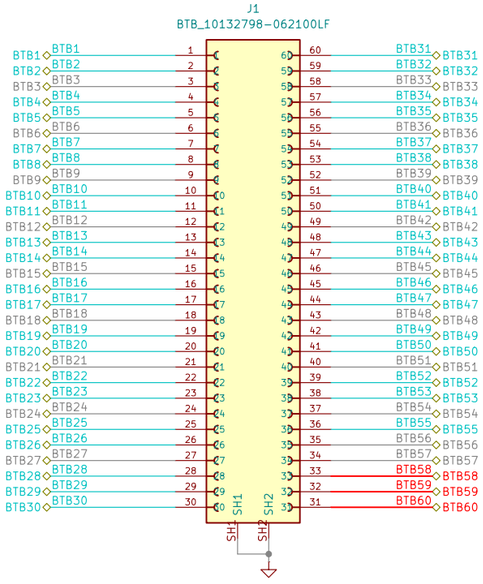
Sheet: /Memory/

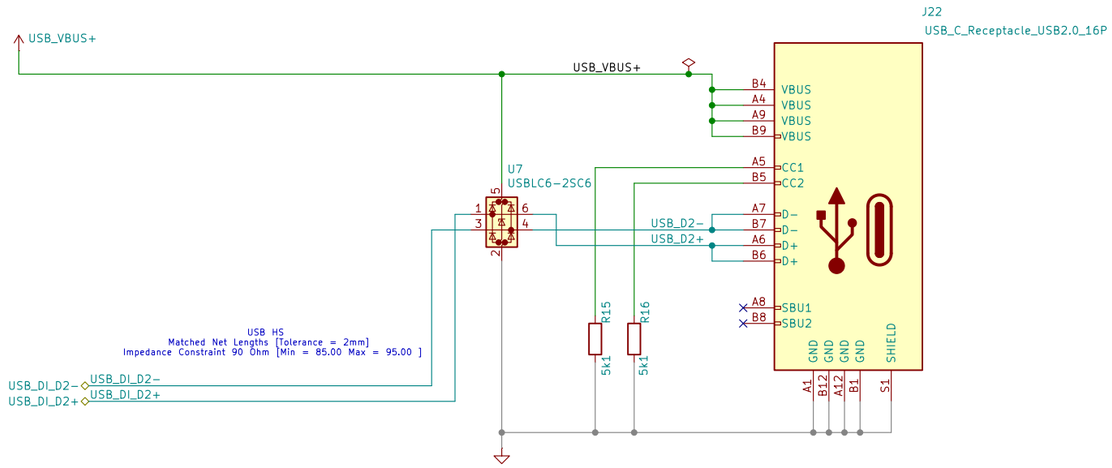
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Title: STM32H7S3 Board

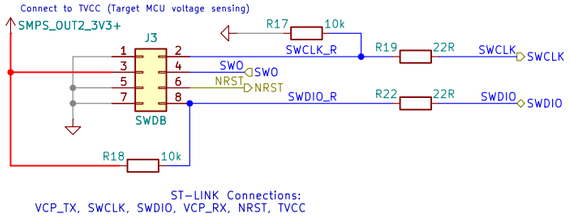
Size: A3	Date: 2026-03-17	**Rev: B**
KiCad E.D.A. 10.0.3		Id: 3/7

Expansion Board Header





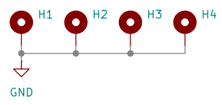
SWD Interface



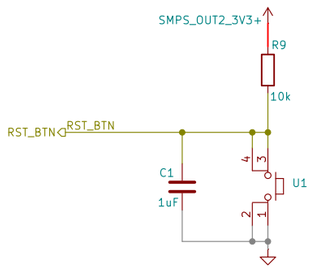
STLINK CONNECTOR
- T_VCC MCU VDDIO (3V3)
- T_SWDIO SWDIO
- T_SWCLK SWCLK
- T_SWO SWO
- T_NRST MCU NRST
- GNDDetect MCU GND *No detection

NOTE:
- GNDDetect Tied to GND by STLINK-V3SET firmware. It might be used by the target to detect the tool. Detection might happen through pull-up on the target board which would be HIGH when STLINK disconnected and would be LOW (pulled LOW by the tie to GND on STLINK). Connect to a general GPIO.

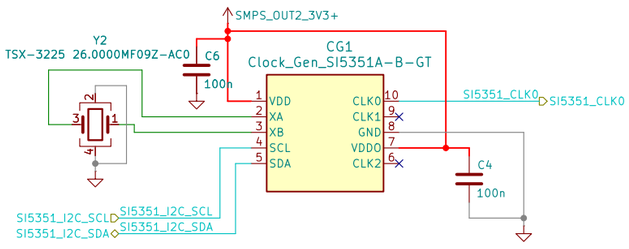
Mounting Holes



Buttons



PSSI INTERAFCE CLOCK



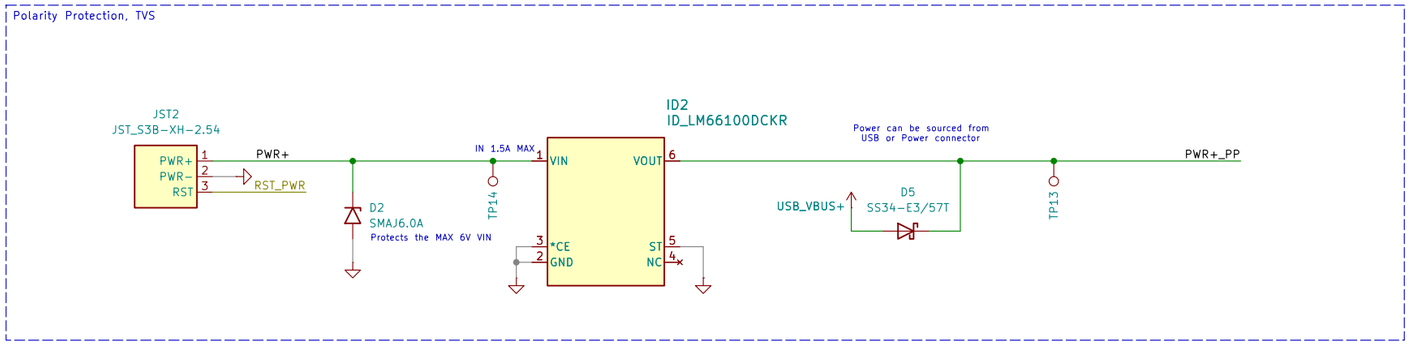
Unused output pins (CLK0-CLK7) should be left floating.

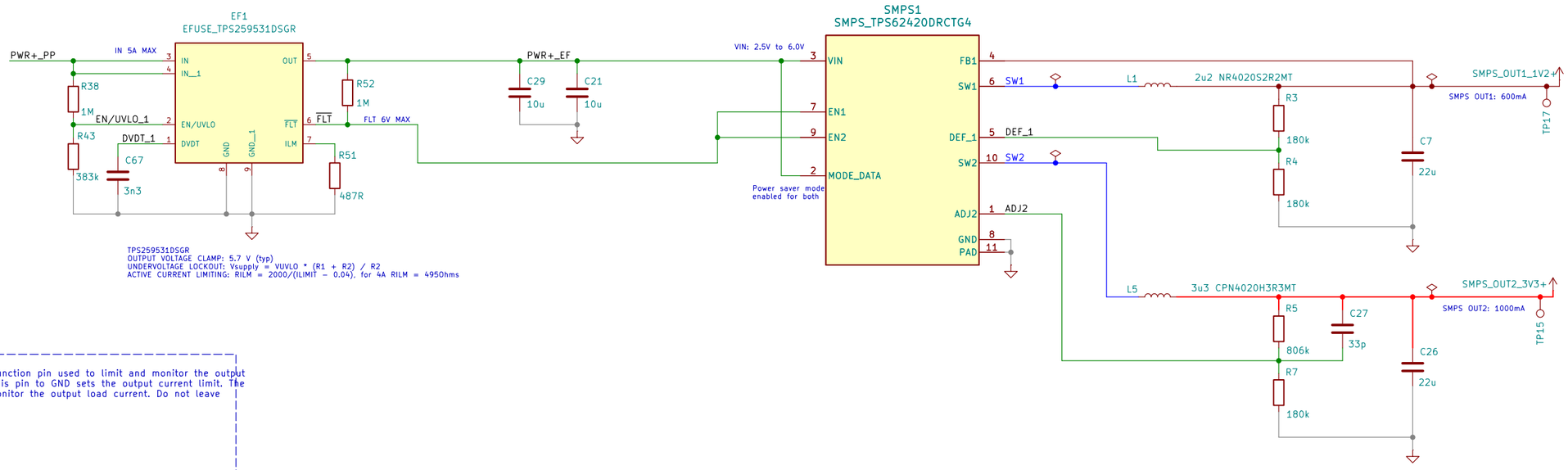
Peter Hancak

Sheet: /Interface_Headers/
File: Interface_Headers.kicad_sch

Title: STM32H7S3 Board

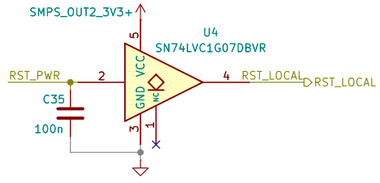
Size: A3	Date: 2026-03-17	Rev: B
KiCad E.D.A. 10.0.3		Id: 4/7



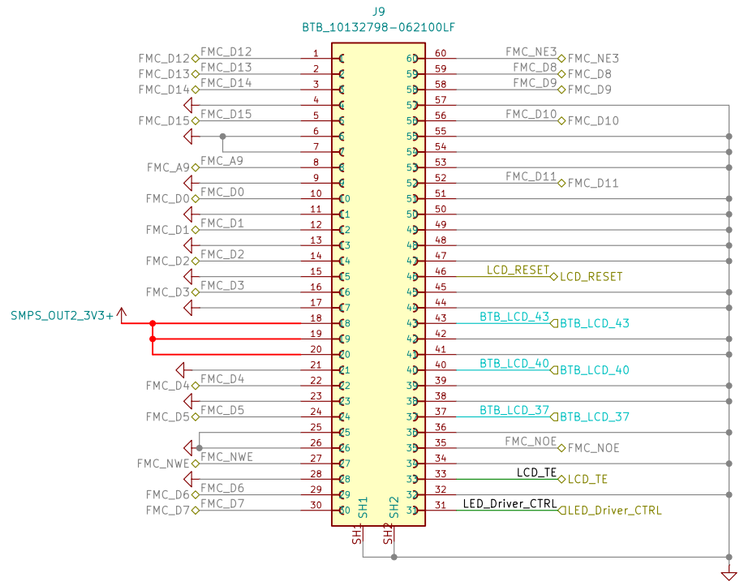


EFUSE TPS259531DSGR NOTES:
– ILM Analog I/O: This is a dual function pin used to limit and monitor the output current. An external resistor from this pin to GND sets the output current limit. The pin voltage can also be used to monitor the output load current. Do not leave floating.

GLOBAL TO LOCAL RST BUFFER



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Sheet: /Power_Management/		
File: Power_Management.kicad_sch		
Title: STM32H7S3 Board		
Size: A3	Date: 2026-03-17	**Rev: B**
KiCad E.D.A. 10.0.3		Id: 5/7



Peter Hancak		
Sheet: /LCD_Interface/		
File: LCD.kicad_sch		
Title: STM32H7S3 Board		
Size: A3	Date: 2026-03-17	**Rev: B**
KiCad E.D.A. 10.0.3		Id: 6/7

POWER MANAGEMENT NOTES:
- The Power Budget Breakdown
  - The LM66100 is rated for a maximum continuous current (I MAX) of 1.5 A across an input voltage range of 1.5 V to 5.5 V. Let's add up the peak consumption of your components on the target 3.3V power domain:
  - STM32H7S3I8T6: Running at its maximum core clock of 600 MHz under a heavy DSP/graphics load, the MCU itself will draw roughly 150 mA to 220 mA.
  - AS4C32M8SA-7TCN (256Mb SDRAM): High-speed continuous burst reading/writing for frame-buffering the LCD draws a maximum peak current of around 90 mA to 120 mA.
  - MX30LF1G18AC-TI (1Gb SLC NAND Flash): During an active page program or block erase sequence, it draws a peak of roughly 30 mA.
  - ILI9488 LCD Module: The controller logic itself only draws a tiny few milliamps, but the LED Backlight panel is power-hungry. At full brightness, a standard 3.5-inch panel draws between 90 mA to 150 mA.
Total Worst-Case Peak Consumption:
Total Current≈220 mA+120 mA+30 mA+150 mA=520 mA

HARDWARE DESIGN CONSIDERATIONS:
- Border/space around the clock IC
- Clock trace isolation
- Impedance matching (USB, SPI, RF)
- Length matching
- Solid GND plane under high speed signals and RF
- Decoupling capacitors
- Power outtput filtering (remove harmonics)
- Stitching vias beside signal vias

PSSI INTERFACE CLOCK: Fly-By (Daisy-Chain) topology
[External Oscillator]->[Sender Pin] (STM32 PSSI_CLK)->[15 cm Trace]->[Receiver Pin]
- The external oscillator drops the clock edge right at the Sender's doorstep.
The Sender senses the clock and immediately dumps data onto the data lines (PSSI_D0-PSSI_D15).
Now, both the Clock and the Data leave the Sender at the exact same moment.
They run down the 15 cm track side-by-side like runners in parallel lanes. Because you will use KiCad's length tuner to make the PSSI_CLK line match the length of the data lines, they will cross the finish line at the Receiver at the exact same instant (T=1.4 ns).
By forcing the clock to travel with the data rather than racing ahead of it, the Receiver gets a perfectly aligned window where the data is stable right when the clock edge hits.

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Sheet: /Notes/
File: Notes.kicad_sch

Title: STM32H7S3 Board

Size: A4	Date: 2026-03-17	**Rev: B**
KiCad E.D.A. 10.0.3		Id: 7/7